

14. The specific latent heat of vaporization of water at 20°C is appreciably greater than the value of 100°C.

This is because

- A** The molecules in the liquid are more tightly bound to one another at 20°C than at 100°C.
- B** More work must be done in expanding the water vapour against atmospheric pressure at 20°C than at 100°C.
- C** The root mean square speed of the vapour molecules is less at 20°C than at 100°C.
- D** The specific latent heat at 20°C includes the energy necessary to raise the temperature of one kilogram of water from 20°C to 100°C.

Answer: A

$$\Delta U = q + w \Rightarrow \Delta U = mL_v + w$$

Consider 1 kg of water, $L_v = \Delta U - w$

Quantity 1: Change in internal energy ΔU

- internal energy is sum of translational k.e. and p.e. of molecules.
- k.e. is proportional to thermodynamic temperature
- p.e. is the separation of molecules (molecules are closer at lower temperature)
- so, increase in internal energy to break the bonds for 20 °C water is higher (temperature remains same so no change in k.e.)

Quantity 2: Work done w

external pressure is the same so work done to expand is same

B: Incorrect. Refer to quantity 2

C: Incorrect. Temperature remains same during vapourisation so no impact

D: Incorrect. Vapourisation can happen at any temperature, no need to increase the temperature.